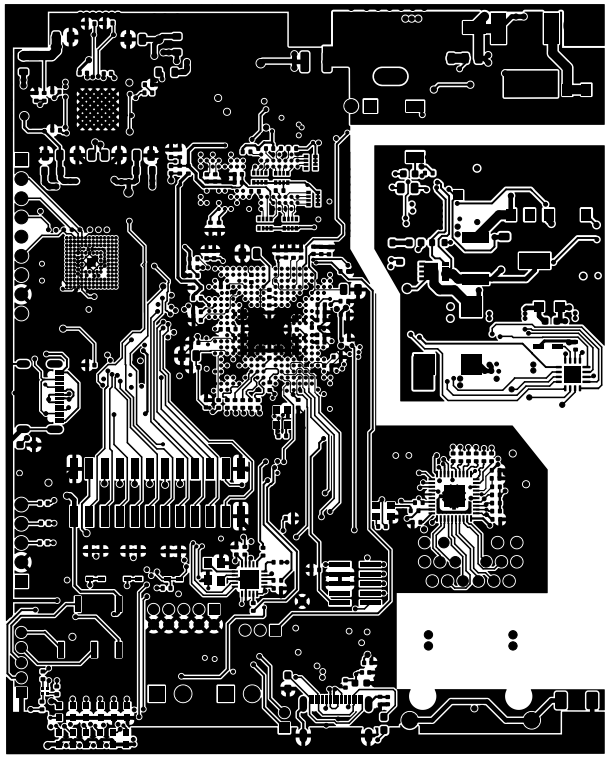
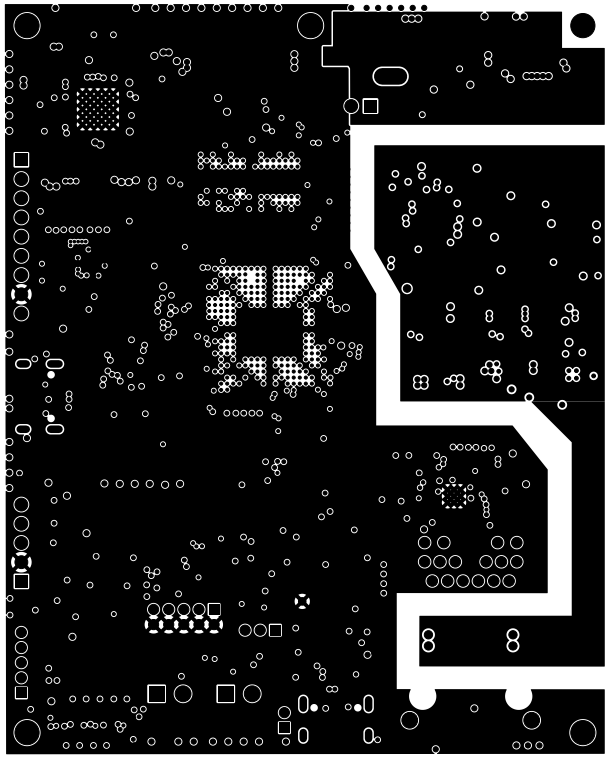
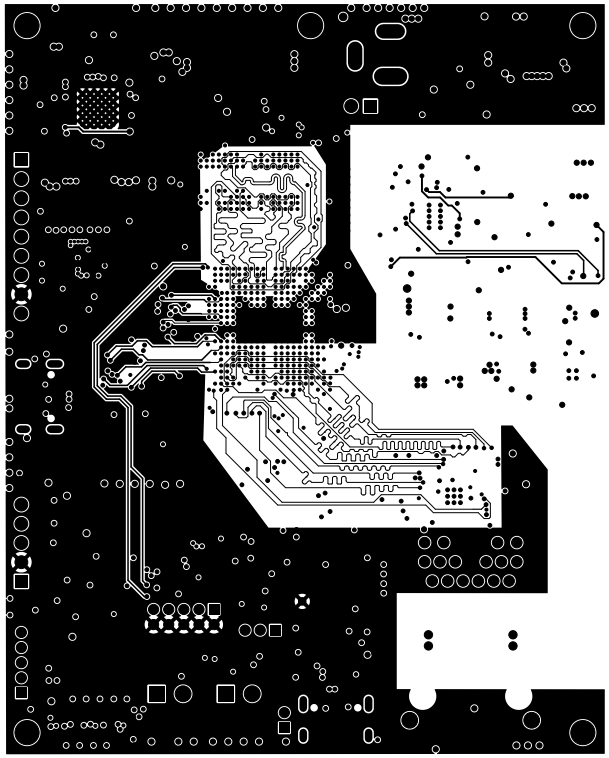
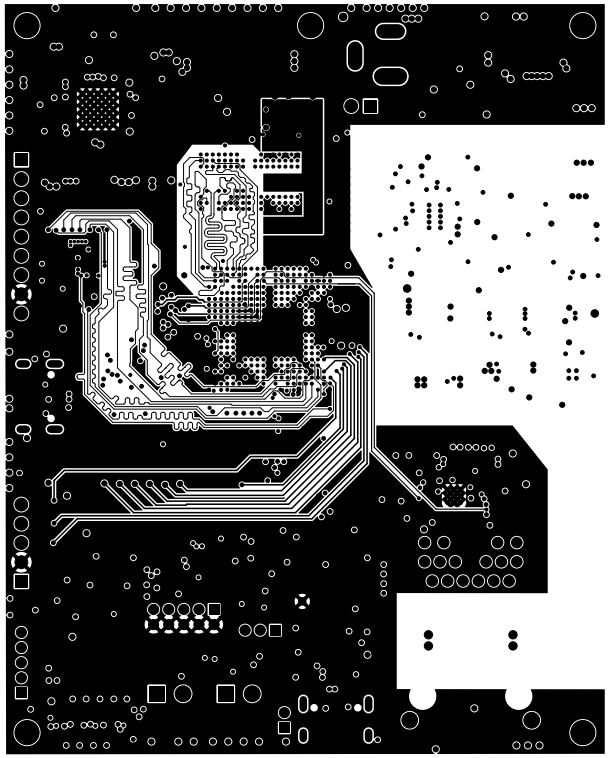
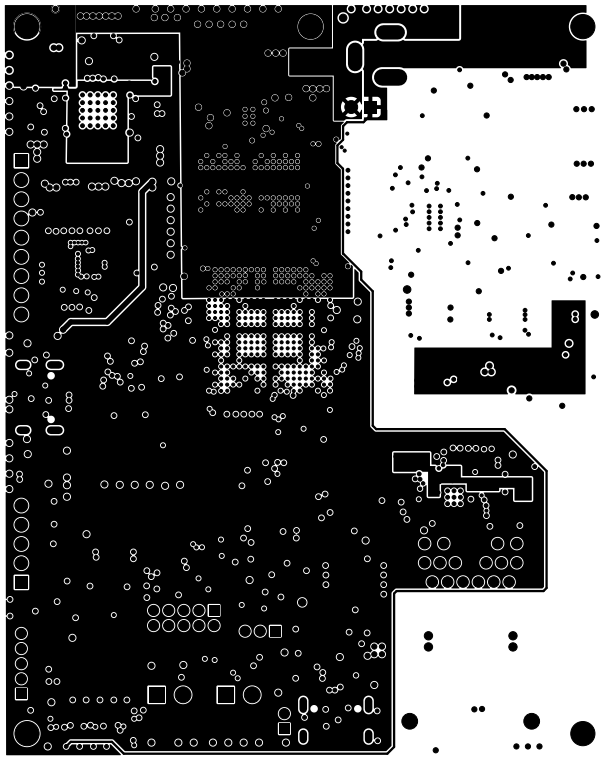
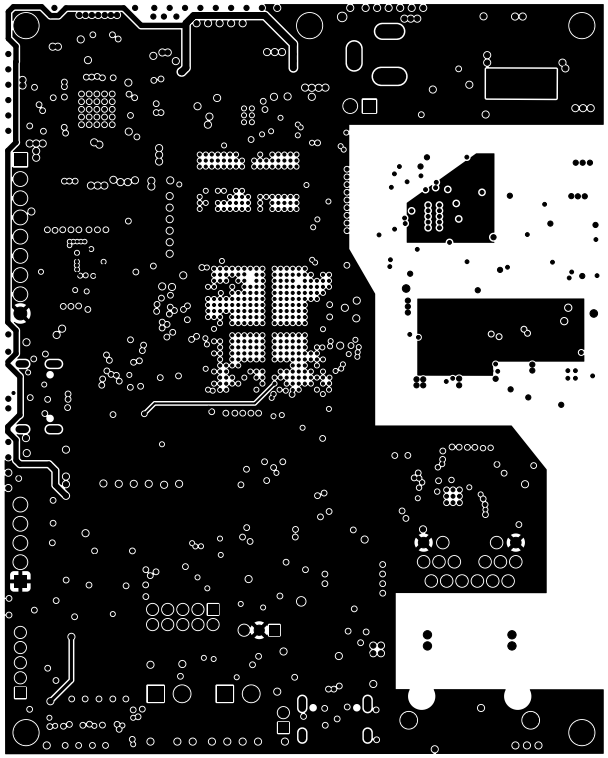


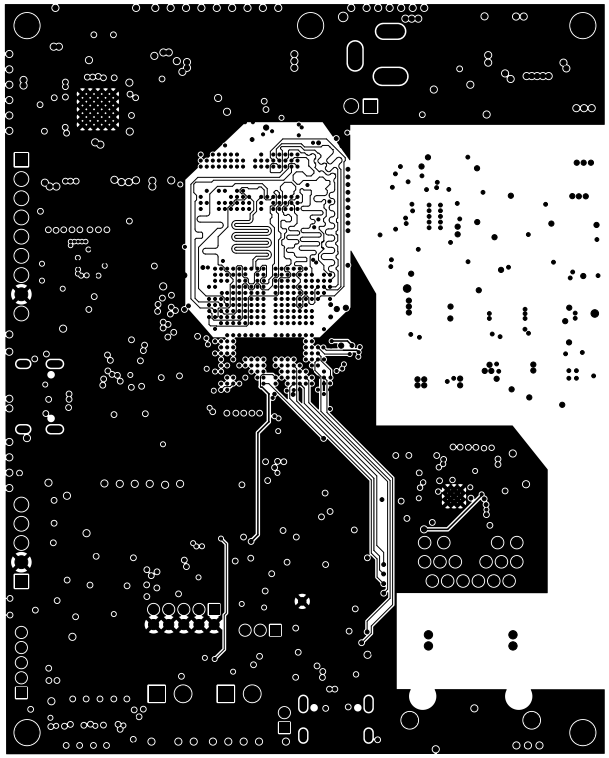


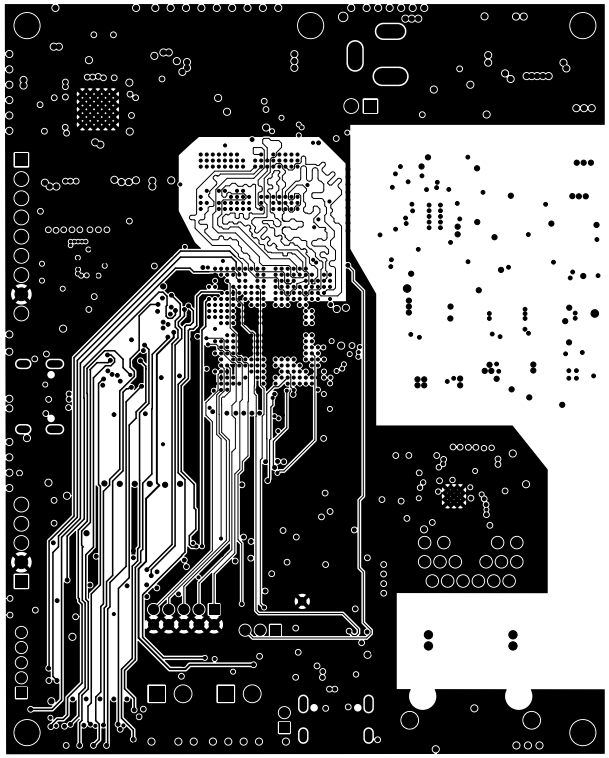


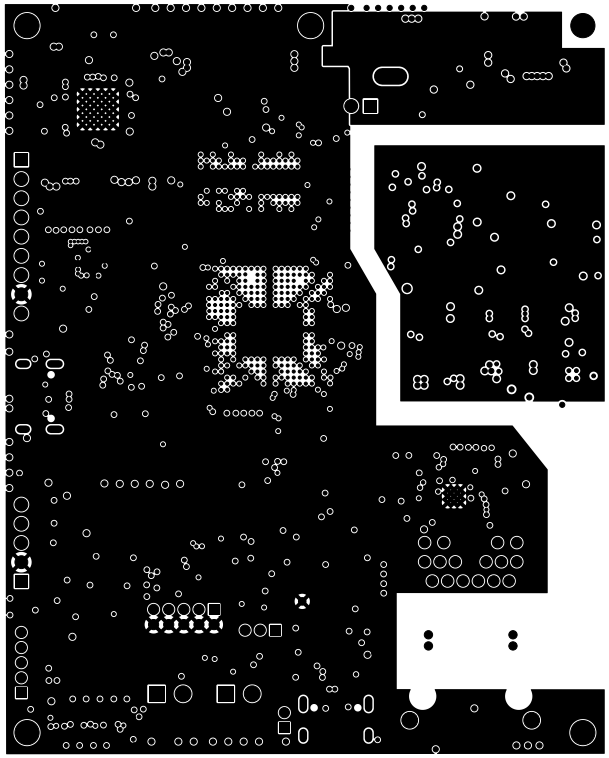


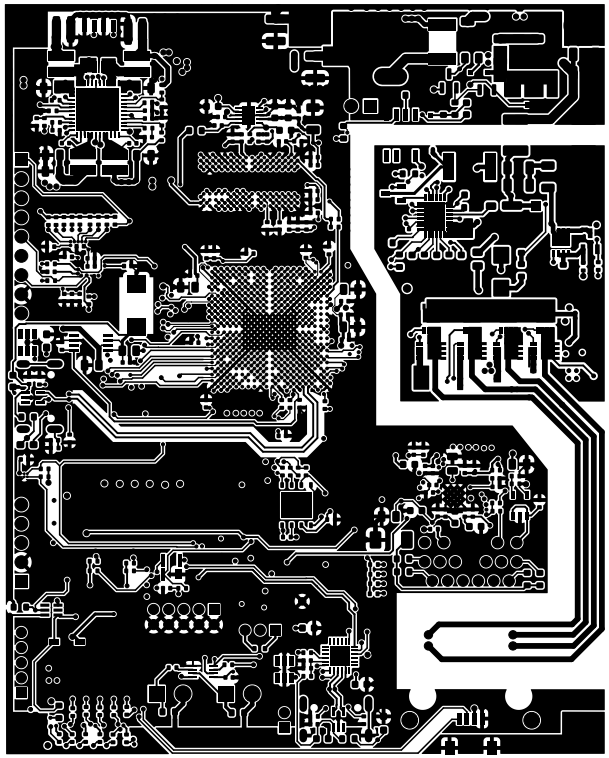


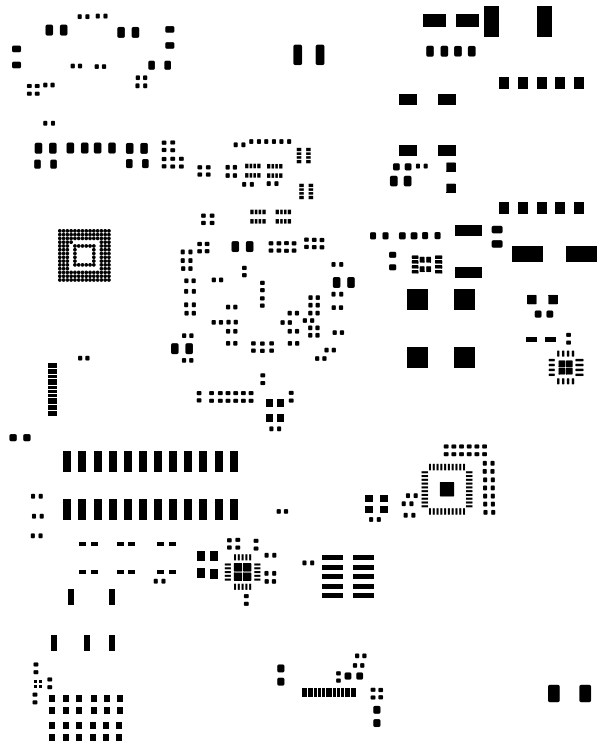


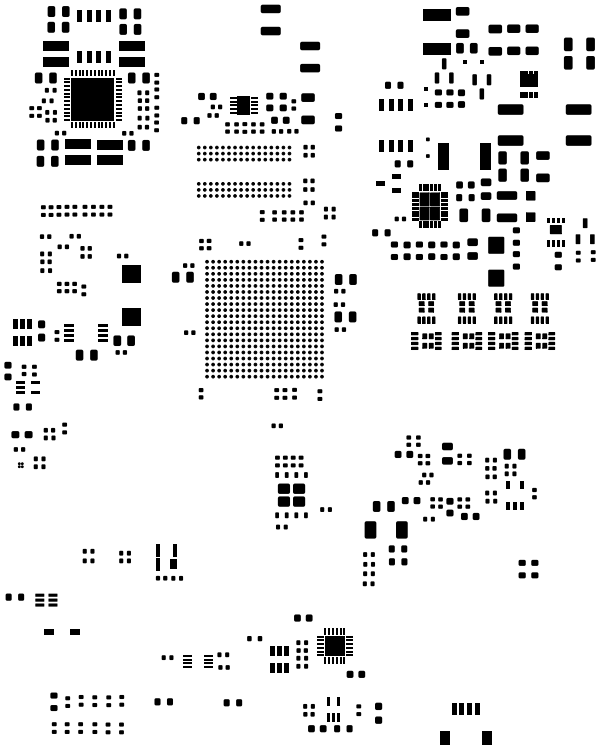


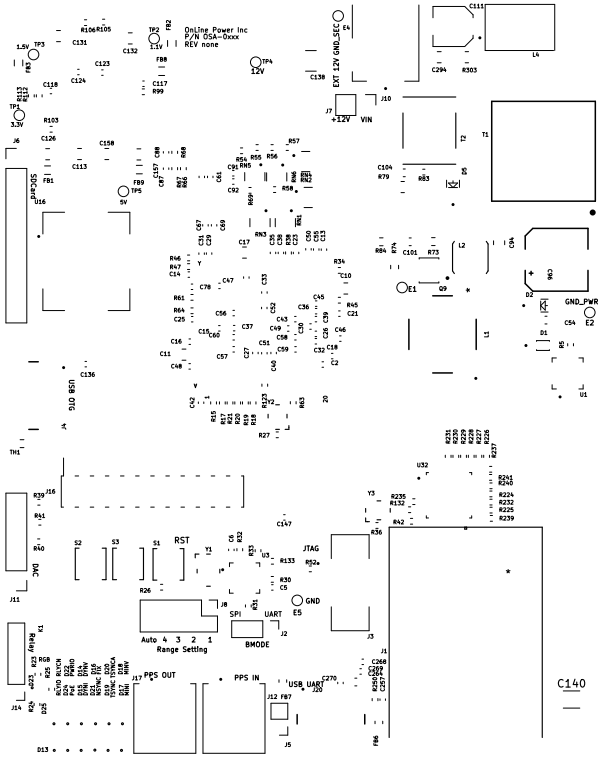




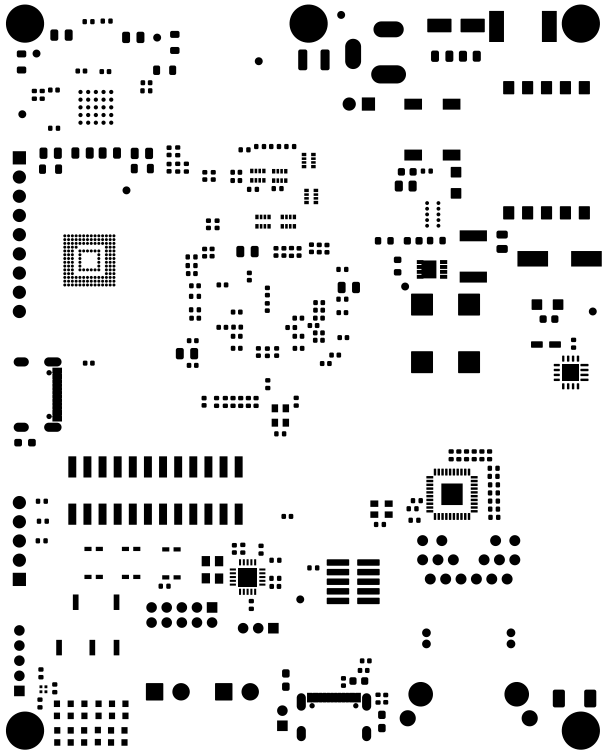


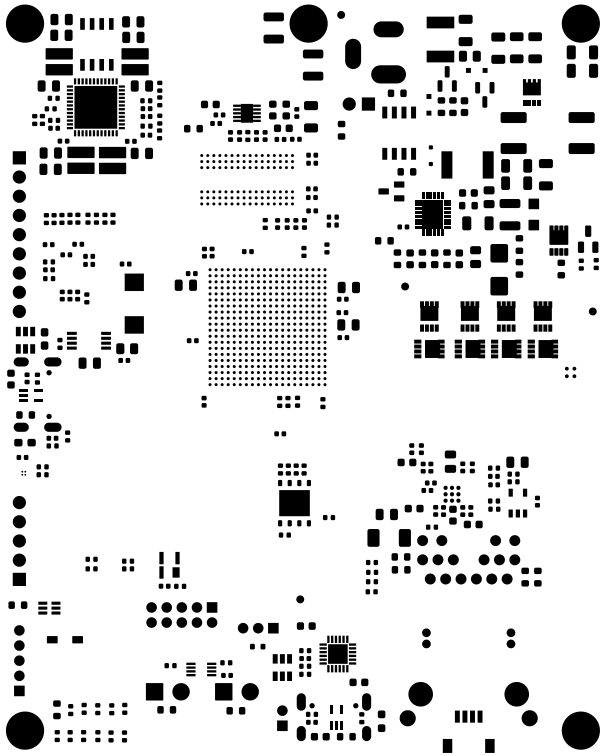





OnLine Power Inc
P/N OSA-Oxxx
REV none
EXT 12V GND_SEC
+12V VIN
GND_PWR
RST
JTAG
SPI
UART
BMODE
Auto 4 3 2 1
Range Setting
PPS OUT
PPS IN
USB UART
C140

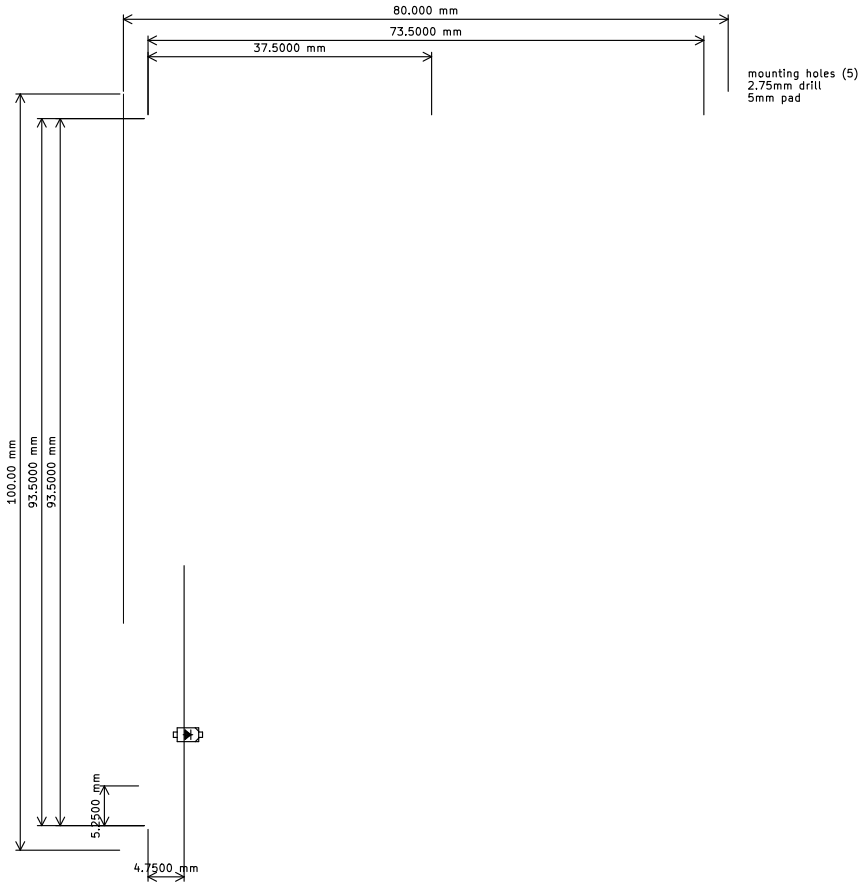
GND
OFED





layer "F.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 1" type "core" Thickness 0.12 mm Material "FR4" EpsilonR 4.7 LossTg 0.02
layer "In1.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 2" type "prepreg" " Thickness 0.08 mm Material "FR4" EpsilonR 4.6 LossTg 0.02
layer "In2.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 3" type "core" Thickness 0.19 mm Material "FR4" EpsilonR 4.5 LossTg 0.02
layer "In3.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 4" type "prepreg" Thickness 0.08 mm Material "FR4" EpsilonR 4.6 LossTg 0.02
layer "In4.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 5" type "core" Thickness 0.19 mm Material "FR4" EpsilonR 4.5 LossTg 0.02
layer "In5.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 6" type "prepreg" Thickness 0.08 mm Material "FR4" EpsilonR 4.6 LossTg 0.02
layer "In6.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 7" type "core" Thickness 0.19 mm Material "FR4" EpsilonR 4.5 LossTg 0.02
layer "In7.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 8" type "prepreg" Thickness 0.08 mm Material "FR4" EpsilonR 4.6 LossTg 0.02
layer "In8.Cu" type "copper" Thickness 0.035 mm
layer "Dielectric 9" type "core" Thickness 0.19 mm Material "FR4" EpsilonR 4.7 LossTg 0.02
layer "B.Cu" type "copper" Thickness 0.035 mm
Finish "None" Option "Impedance Controlled"

all via-in-pad
filled and capped
(IPC-4761 Type VII
plated-over fill)



CONTROLLED IMPEDANCE SIGNAL GROUPS

Signal Group	Nets	Type	Layer(s)	Width	Gap	Target Z	Calculated
USB 2.0	USB0_DP, USB0_DN	Diff	F.Cu/B.Cu	10mil	12mil	90Ω	90.235Ω
Ethernet MDI (PHY side)	MD1± - MD4± (8 nets)	Diff	F.Cu	8mil	9mil	100Ω	100.052Ω
DDR3 CK/DQS (coupled)	DMC0_CKP/N, LDQSP/N, UDQSP/N	Diff	In2/In3/In6	5mil	10mil	80Ω	75.2Ω
DDR3 addr/cmd/data (SE)	DMC0 addr/cmd/data bus (36 nets)	SE	In2/In3/In6	5mil	–	40Ω	36.6Ω
eMMC (U16)	see U16 via-in-pad notes	SE	–	–	–	50Ω	not re-verified

